

Department of Computer Science and Engineering

CSE 2384

Object-Oriented Programming Lab

Assignment

Adventure Guild Management System

Total Marks: 100

Group Size: 3-4 Students

Course Outcome Assessed

CO2: Analyze problems and select appropriate object-oriented constructs for software development.

Assignment Objective

The objective of this assignment is to evaluate students' ability to implement a software system using object-oriented programming concepts and techniques. Students are expected to transform a software design into a functioning Python program by selecting and applying appropriate object-oriented constructs.

Project Scenario: Adventure Guild Management System

The continent of Tamriel is home to hundreds of adventurers who explore dangerous dungeons, complete quests, defeat monsters, and gather valuable treasures. To organize these activities, adventurers join guilds that manage members, quests, rewards, inventories, and rankings.

To improve efficiency and reduce manual record keeping, the Guild Council has decided to develop an **Adventure Guild Management System**.

The software should support the following activities:

- Register and manage adventurers.
- Organize guild information.
- Create and manage quests.
- Assign quests to adventurers.
- Track quest completion.
- Manage inventories and equipment.
- Record rewards earned by adventurers.
- Maintain guild rankings and member statistics.

Your task is to implement this system in Python using object-oriented programming principles.

Implementation Requirements

Your program must satisfy the following minimum requirements.

- At least **10 classes**.

- Each class must contain at least **3 attributes**.
- Each class must contain at least **3 meaningful methods**.
- At least **one inheritance hierarchy**.
- The inheritance hierarchy must contain at least **one parent class and three child classes**.
- At least **two composition relationships**.
- At least **three association relationships**.
- At least **one example of polymorphism**.
- A `main()` function must be provided to demonstrate the system.

Program Demonstration Requirements

The `main()` function must demonstrate the following:

1. Creation of guilds.
2. Creation of adventurers.
3. Creation of quests.
4. Assignment of quests.
5. Completion of quests.
6. Inventory management.
7. Reward distribution.
8. Display of member statistics or rankings.
9. Demonstration of inheritance.
10. Demonstration of polymorphism.

Students may implement additional features if desired.

Submission Files

Students must submit the following files.

File 1: Python Source Code

Submit a single Python file.

Example:

`group01_adventure_guild.py`

Requirements:

- All classes must be contained in the same file.
- The program must execute without modification.
- The program must contain a `main()` function.
- The program must not rely on external libraries.

File 2: PDF Report

Submit a PDF report containing the following sections.

Report Structure

Cover Page

Include:

- Assignment Title
- Course Code
- Course Title
- Group Members
- Student IDs

Section 1: Implemented Classes

Provide a summary of all implemented classes.

Suggested format:

Class	Purpose
Guild	Manages guild members and quests
Adventurer	Represents a guild member
Quest	Represents quests available to members

Section 2: OOP Features Used

Briefly explain how the following concepts were implemented:

- Inheritance
- Composition
- Association
- Polymorphism

Students should reference actual classes from their implementation.

Section 3: Program Output

Include screenshots of program execution demonstrating:

- Object creation
- Quest assignment
- Inventory management
- Reward distribution
- Polymorphic behavior

Section 4: Reflection

Answer the following questions:

1. What was the most difficult class to implement?
2. Which object-oriented concept was most useful?
3. What would you improve if additional time were available?

Mark Distribution

Component	Marks
Class Implementation	30
Attributes and Methods Design	20
Inheritance and Polymorphism	20
Composition and Association	10
Program Demonstration (<code>main()</code>)	10
PDF Report	10
Total	100

Important Notes

- The implementation should reflect the software design developed in CSE 2383.
- The focus of this assignment is object-oriented programming, not user interface design.
- Students should prioritize meaningful class design and interactions.
- Marks will be awarded based on correct use of object-oriented constructs rather than program size.
- Every class should serve a meaningful purpose within the system.
- During evaluation, students may be asked to explain their code and design decisions.